Machine Learning & AI Laboratory Manual

Department: AI & Data Science | **Semester:** 6 | **Course Code:** AD601L

Academic Year: 2026-2027 | **University Autonomous Syllabus**

Experiment 1: Linear Regression & Gradient Descent Optimization

Objective: Implement univariate linear regression from scratch using NumPy and visualize the cost function convergence.

Mathematical Formulation:
Hypothesis: h_theta(x) = theta_0 + theta_1 * x
Cost Function (MSE): J(theta) = (1 / 2m) * sum((h_theta(x_i) - y_i)^2)
Weight Update: theta_j := theta_j - alpha * (1/m) * sum((h_theta(x_i) - y_i) * x_i_j)

```
# Python Implementation snippet:
import numpy as np
def gradient_descent(X, y, lr=0.01, epochs=1000):
    m, n = X.shape
    theta = np.zeros(n)
    for _ in range(epochs):
        errors = np.dot(X, theta) - y
        theta -= (lr / m) * np.dot(X.T, errors)
    return theta
```

Experiment 2: K-Means Clustering for Customer Segmentation

Algorithm Steps:
1. Randomly initialize K centroids from the dataset.
2. Assign each observation to the nearest centroid using Euclidean distance: d(x, c) = sqrt(sum((x_i - c_i)^2)).
3. Recalculate centroids as the arithmetic mean of all points assigned to that cluster.
4. Repeat until centroid movement drops below threshold epsilon.

Lab Viva Questions:
1. How does learning rate affect gradient descent convergence?
2. Explain the Elbow method for choosing the optimal value of K.
3. What is the difference between L1 (Lasso) and L2 (Ridge) regularization?